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MASTER'S DEGREE IN PHYSICS: RADIATION, NANOTECHNOLOGY, PARTICLES AND ASTROPHYSICS

MATHEMATICAL AND NUMERICAL COMPLEMENTS

DIFFUSION MONTE CARLO AND THE HARMONIC ATOM

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Abstract

The ground-state energy of a system of bosonic particles in a harmonic potential is studied numerically as a function of the interaction parameter β^2 and the number of particles N, using the Diffusion Monte Carlo (DMC) method. Both the basic formulation without importance sampling and its guided version using the Trotter approximation are considered.

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1 Introduction

Starting from the time-dependent Schrödinger equation with a reference energy E_T

$$i\hbar \frac{\partial \Psi(\mathbf{r}, t)}{\partial t} = (\hat{H} - E_T)\Psi(\mathbf{r}, t), \tag{1.1}$$

and passing to imaginary time $\tau = it/\hbar$, one obtains

$$\frac{\partial \Psi(\mathbf{r}, \tau)}{\partial \tau} = -(\hat{H} - E_T)\Psi(\mathbf{r}, \tau) \tag{1.2}$$

where $\mathbf{r}$ is the position vector of the particles in configuration space, $\mathbf{r} \in \Omega$, $\mathbf{r} = (\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_\mathbf{N})$. Equation 1.2 has the *symbolic* solution

$$\Psi(\mathbf{r}, \tau) = e^{-(\hat{H} - E_T)\tau}\Psi(\mathbf{r}, \tau = 0).$$

Assuming the initial wavefunction can be expanded in eigenfunctions of the Hamiltonian, $\hat{H}\psi_n = E_n\psi_n$, with energies E_n in increasing order ($E_1 < E_2 < \cdots < E_n$),

$$\Psi(\mathbf{r}, \tau) = \sum_n e^{-(E_n - E_T)\tau} \langle \psi_n | \Psi_0 \rangle \psi_n.$$

At large imaginary times the ground state dominates, so

$$\lim_{\tau \to \infty} \Psi(\mathbf{r}, \tau) \to e^{-(E_0 - E_T)\tau} \langle \psi_0 | \Psi_0 \rangle \psi_0.$$

The main difficulty in applying the time-evolution operator is computing the exponential of the Hamiltonian, since the kinetic $\hat{T}$ and potential $\hat{V}$ operators do not generally commute:

$$|\Psi(\mathbf{r}, \tau)\rangle = e^{-(\hat{H} - E_T)\tau} |\Psi(\mathbf{r}, 0)\rangle .$$

The *short-time approximation* is therefore applied, computing the state at each small time step $\Delta\tau$:

$$|\Psi(\mathbf{r}, \tau + \Delta\tau)\rangle = e^{-(\hat{H} - E_T)\Delta\tau} |\Psi(\mathbf{r}, \tau)\rangle .$$

Applying the Trotter approximation to the time-evolution operator [3]:

$$e^{-(\hat{H} - E_T)\Delta\tau} = e^{-(\hat{V} - E_T)\Delta\tau/2} e^{-\hat{T}\Delta\tau} e^{-(\hat{V} - E_T)\Delta\tau/2} + \mathcal{O}(\Delta\tau^3). \tag{1.3}$$

In the position representation the state at time $\tau + \Delta\tau$ is

$$\Psi(\mathbf{r}, \tau + \Delta\tau) = \langle \mathbf{r} | \Psi(\tau + \Delta\tau) \rangle = \langle \mathbf{r} | e^{-(\hat{H} - E_T)\Delta\tau} | \Psi(\tau) \rangle .$$

Using the resolution of identity in $\mathbf{r}'$,

$$\Psi(\mathbf{r}, \tau + \Delta\tau) = \int_{\mathbf{r}'} \langle \mathbf{r} | e^{-(\hat{H} - E_T)\Delta\tau} | \mathbf{r}' \rangle \Psi(\mathbf{r}', \tau) d\mathbf{r}'.$$

Inserting resolutions of identity in $\mathbf{r}''$ and $\mathbf{r}'''$ with the approximation 1.3,

$$\Psi(\mathbf{r}, \tau + \Delta\tau) = \int_{\mathbf{r}'} \int_{\mathbf{r}''} \int_{\mathbf{r}'''} \langle \mathbf{r} | e^{-(\hat{V} - E_T)\Delta\tau/2} | \mathbf{r}'' \rangle$$
$$\cdot \langle \mathbf{r}'' | e^{-\hat{T}\Delta\tau} | \mathbf{r}''' \rangle \langle \mathbf{r}''' | e^{-(\hat{V} - E_T)\Delta\tau/2} | \mathbf{r}' \rangle \Psi(\mathbf{r}', \tau) \, d\mathbf{r}' d\mathbf{r}'' d\mathbf{r}'''.$$

Computing each factor individually:

- $\langle \mathbf{r} | e^{-(\hat{V} - E_T)\Delta\tau/2} | \mathbf{r}'' \rangle = e^{-(\hat{V}(\mathbf{r}'') - E_T)\Delta\tau/2} \delta(\mathbf{r} - \mathbf{r}'')$,
- $\langle \mathbf{r}''' | e^{-(\hat{V} - E_T)\Delta\tau/2} | \mathbf{r}' \rangle = e^{-(\hat{V}(\mathbf{r}') - E_T)\Delta\tau/2} \delta(\mathbf{r}''' - \mathbf{r}')$,
- The kinetic energy operator is $-\Delta\tau\hat{T} = \frac{\Delta\tau\hbar^2}{2m}\nabla^2$, so:
  $\langle \mathbf{r}'' | e^{-\hat{T}\Delta\tau} | \mathbf{r}''' \rangle = \dfrac{1}{(4\pi D\Delta\tau)^{N/2}} \exp\left(-\dfrac{(\mathbf{r}'' - \mathbf{r}''')^2}{4\pi D\Delta\tau} \right)$
  where $D = \hbar^2/2m$ and $\sigma^2 = 2D\Delta\tau$.

The total state evolution is therefore

$$\Psi(\mathbf{r}, \tau + \Delta\tau) = \frac{1}{(2\pi\sigma^2)^{N/2}} \int_{\mathbf{r}'} e^{-\frac{(\mathbf{r} - \mathbf{r}')^2}{2\sigma^2}} e^{-\Delta\tau/2\left[\hat{V}(\mathbf{r}) + \hat{V}(\mathbf{r}') - 2E_T\right]} \Psi(\mathbf{r}', \tau) \, d\mathbf{r}'. \tag{1.4}$$

1.1 Diffusion Monte Carlo

In practice, when the number of particles N is large the integral in 1.4 is impossible to evaluate directly. A stochastic interpretation based on Monte Carlo methods, known as *Diffusion Monte Carlo* (DMC) [2], is therefore employed.

In the DMC method the wavefunction is not represented as a continuous function but by a population of M fictitious particles called *walkers*. Each walker corresponds to a point in the configuration space of the system, i.e. we write Ψ as

$$\Psi(\mathbf{r}, \tau) = \sum_{n=0}^{M} \delta(\mathbf{r} - \mathbf{r_i}(\tau)).$$

The density of walkers in configuration space is proportional to the value of the wavefunction. The evolution equation takes the general form

$$\text{new state} = \text{propagator} \times \text{old state}.$$

In the Monte Carlo context the propagator is interpreted as a probability distribution governing the transition of a walker from its current position to a new one. The propagator separates into two main contributions: diffusion and branching.

The Gaussian term of the propagator describes a diffusion process. At each time step each walker moves randomly according to a normal distribution — the new position is obtained by adding a Gaussian random displacement to the previous one. This represents the effect of the kinetic energy and can be interpreted as a random walk through configuration space.

The potential-dependent term does not directly modify the walker's position but its statistical weight[1]. This factor determines the probability of survival or reproduction of each walker. When the local potential is low, the walker is more likely to be replicated. In regions of high potential, walkers tend to disappear; the process thus acts as a natural-selection mechanism that favours low-energy regions[2].

The Langevin equation associated with the Fokker–Planck equation 1.2 is [5]

$$d\mathbf{r}(\tau) = \sqrt{2D}\, d\mathbf{W}(\tau),$$

where $d\mathbf{W}$ is a Wiener process. In discrete form the process is Markovian and the walker positions obey

$$\mathbf{r_{n+1}} = \mathbf{r_n} + \sqrt{2D\Delta\tau}\, \boldsymbol{\eta_n} \tag{1.5}$$

where $\boldsymbol{\eta_n} \sim \mathcal{N}(0,1)$, and at each time step walkers are multiplied by the weight $w_n = \exp\left[-\Delta\tau/2 \cdot (V(\mathbf{r_n}) + V(\mathbf{r_{n+1}}) - 2E_T)\right]$.

1.1.1 Energy estimation

Two approaches exist for estimating the ground-state energy from E_T. Integrating expression 1.4 over all walker positions $\mathbf{r}$ gives an estimate of the population $M(\tau)$:

$$M(\tau + \Delta\tau) = \int_{\mathbf{r}} \Psi(\mathbf{r}, \tau + \Delta\tau)\, d\mathbf{r} = e^{-(E_0 - E_T)\Delta\tau} M(\tau).$$

The first option is to keep a population of walkers that evolves with time (via the weights) and hold E_T constant, so iterating the relation above gives

$$E_0 \approx E_T - \frac{1}{\tau} \ln \frac{M(\tau)}{M(0)},$$

or equivalently by averaging several measurements of

$$E_0 \approx E_T + \frac{1}{\Delta\tau} \ln \frac{M(\tau)}{M(\tau + \Delta\tau)}.$$

[1] In the simulation the multiplicative factor m_i is taken as the integer part of the weight w_i plus a uniform random number between 0 and 1 to avoid bias.

[2] For potentials that diverge to $-\infty$, walkers reproduce indefinitely and a special treatment is required in DMC.

In this scheme the population at the next time step is the previous population with each walker weighted by its statistical weight, which requires a population-control method to prevent runaway growth. The alternative is to hold the population constant at M_0 and let $E_T(\tau)$ vary:

$$\frac{dM(\tau)}{d\tau} = \int_{\mathbf{r}} \frac{\partial \Psi(\mathbf{r},\tau)}{\partial \tau}\, d\mathbf{r} = \int_{\mathbf{r}} (\hat{T} + \hat{V} - E_T(\tau))\Psi(\mathbf{r},\tau)\, d\mathbf{r} = \int_{\mathbf{r}} (E_0 - E_T(\tau))\Psi(\mathbf{r},\tau)\, d\mathbf{r} = 0.$$

The kinetic term $\int_{\mathbf{r}} \nabla \cdot (\nabla \Psi(\mathbf{r},\tau))\, d\mathbf{r}$ is evaluated at the boundaries of configuration space; for a wavefunction that decays sufficiently fast it can be neglected. This is valid for the harmonic potential considered here.

The condition on E_T is

$$E_T(\tau) = \frac{\int \hat{V}(\mathbf{r})\Psi(\mathbf{r},\tau)\, d\mathbf{r}}{\int \Psi(\mathbf{r},\tau)\, d\mathbf{r}} = \langle \hat{V} \rangle_{\Psi(\tau)},$$

adjusted dynamically at each time step as

$$E_T(\tau + \Delta\tau) = \langle \hat{V} \rangle_{\Psi(\tau)}.$$

The ground-state energy E_0 is estimated as the time average of $E_T(\tau)$.

To avoid both the runaway-growth problem and the fixed-population bias, a mixed scheme is used in which the population is allowed to *oscillate* around a target value M_0:

$$E_T(\tau + \Delta\tau) = \langle \hat{V} \rangle_{\Psi(\tau)} + \frac{\gamma}{\Delta\tau} \ln\left(\frac{M_0}{M(\tau)}\right),$$

where γ is a small damping factor. If $M(\tau)$ falls below M_0, the term becomes positive, raising E_T and encouraging walker replication; if the population exceeds M_0, E_T decreases and walkers are annihilated.

1.2 Importance Sampling

Although the DMC method largely overcomes the curse of dimensionality, it introduces other problems. Walkers explore configuration space by free diffusion, frequently visiting regions where the wavefunction is nearly zero and wasting computational effort on regions irrelevant to the ground state. This causes the statistical variance to grow rapidly, leading to large fluctuations.

To improve efficiency, a trial (guide) wavefunction $\Psi_T(\mathbf{r})$ that approximates the ground state is introduced. Instead of sampling Ψ directly, one works with the function

$$f(\mathbf{r},\tau) = \Psi_T(\mathbf{r})\Psi(\mathbf{r},\tau). \tag{1.6}$$

Substituting this definition into the evolution equation 1.2 yields a Fokker–Planck equation

$$\frac{\partial f}{\partial \tau} = D\nabla^2 f - \nabla \cdot (\mathbf{F}(\mathbf{r}) f) - [E_L(\mathbf{r}) - E_T]\, f, \tag{1.7}$$

where $\mathbf{F}(\mathbf{r}) = 2D\nabla \ln \Psi_T(\mathbf{r})$ is the drift force and $E_L(\mathbf{r}) = \frac{\hat{H}\Psi_T(\mathbf{r})}{\Psi_T(\mathbf{r})}$ is the local energy. The kinetic term now becomes a drift-diffusion process and the branching term depends only on the local energy. The Trotter approximation for the state evolution reads

$$\begin{aligned} f(\mathbf{r}, \tau + \Delta\tau) = \int_{\mathbf{r}'} \frac{1}{(2\pi\sigma^2)^{N/2}} \exp\left(-\frac{1}{2\sigma^2}\left[\mathbf{r} - \mathbf{r}' - \mathbf{F}(\mathbf{r}')\Delta\tau\right]^2\right) \\ \cdot \exp\left(\Delta\tau\left[E_T - \frac{1}{2}(E_L(\mathbf{r}) + E_L(\mathbf{r}'))\right]\right) f(\mathbf{r}', \tau)\, d\mathbf{r}'. \end{aligned} \tag{1.8}$$

With this transformation walkers no longer perform a pure random walk; instead they *feel* a drift towards regions where Ψ_T is large. Sampling is concentrated in the physically relevant regions of configuration space, the statistical variance is drastically reduced, and the branching process depends on the local energy rather than the bare potential.

The discrete Langevin equation guiding the walkers is

$$\mathbf{r}_{n+1} = \mathbf{r}_n + \mathbf{F}(\mathbf{r}_n)\Delta\tau + \sqrt{2D\Delta\tau}\,\boldsymbol{\eta}_n,$$

and walkers are multiplied by the weights $w_n = \exp\left[-\Delta\tau/2\left(E_L(\mathbf{r}_n) + E_L(\mathbf{r}_{n+1}) - 2E_T\right)\right]$.

Finally, the ground-state energy E_0 is estimated as the mean local energy with respect to f. Using the Hermiticity of the Hamiltonian:

$$\begin{aligned} \langle E_L \rangle_f &= \frac{\int f E_L\, dr}{\int f\, dr} = \frac{\int \Psi_T \Psi_0 \frac{\hat{H}\Psi_T}{\Psi_T}\, dr}{\int \Psi_T \Psi_0\, dr} = \frac{\int \Psi_0 \hat{H} \Psi_T\, dr}{\int \Psi_T \Psi_0\, dr} \\ &= \frac{\int \Psi_T \hat{H} \Psi_0\, dr}{\int \Psi_T \Psi_0\, dr} = \frac{\int \Psi_T E_0 \Psi_0\, dr}{\int \Psi_T \Psi_0\, dr} = E_0. \end{aligned}$$

The population-control update is

$$E_T(\tau + \Delta\tau) = \langle E_L \rangle_{f(\tau)} + \frac{\gamma}{\Delta\tau} \ln\left(\frac{M_0}{M(\tau)}\right).$$

2 The Harmonic Atom

We consider particles moving in three-dimensional space and interacting through the Hamiltonian

$$\hat{H} = -\frac{1}{2}\sum_{k=1}^{N} \nabla^2_{\mathbf{r}_k} + \frac{1}{2}\sum_{k=1}^{N} \mathbf{r}_k^2 - \frac{\beta^2}{2N}\sum_{l>k=1}^{N} |\mathbf{r}_k - \mathbf{r}_l|^2, \tag{2.1}$$

with $0 \leq \beta^2 < 1$. The analytical ground-state energy is [1]

$$E_{\text{exact}} = \frac{3}{2}\left[1 + (N-1)\sqrt{1-\beta^2}\right]. \tag{2.2}$$

2.1 Pure Diffusion Monte Carlo

Since the potential is harmonic, walkers are initialised from a Gaussian centred at the origin, $\mathbf{r}_i(0) \sim \mathcal{N}(0, D)$. Because this initialisation is arbitrary and may not represent the true ground state, the first steps are discarded — a procedure known as thermalisation or *burn-in* [4]. Here the thermalisation time is set to `0.3*Nsteps`, where *Nsteps* is the total number of time steps $\Delta\tau$.

At each time step $\Delta\tau$ walkers evolve by free diffusion via the Langevin equation $\mathbf{r_{n+1}} = \mathbf{r_n} + \sqrt{2D\Delta\tau}\,\boldsymbol{\eta_n}$ and are multiplied by the weight $w_n = \exp\left[-\Delta\tau/2 \cdot (V(\mathbf{r_n}) + V(\mathbf{r_{n+1}})\right.$

The total number of walkers is regulated dynamically by adjusting E_T via $E_T(\tau + \Delta\tau) = \langle \hat{V} \rangle_{\Psi(\tau)} + \frac{1}{\Delta\tau} \ln \frac{M_0}{M(\tau)}$, where M_0 is the target population. Once the stationary regime is reached, the time average of E_T provides an estimate of the ground-state energy.

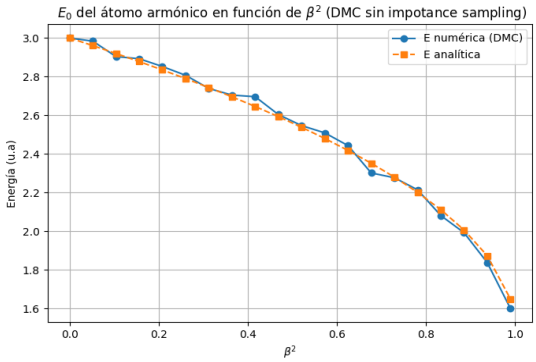
2.1.1 Results

Figure 1 shows the numerical results obtained with the pure DMC method described in section 1.1, together with the analytical solution equation 2.2, for several values of the particle number N.

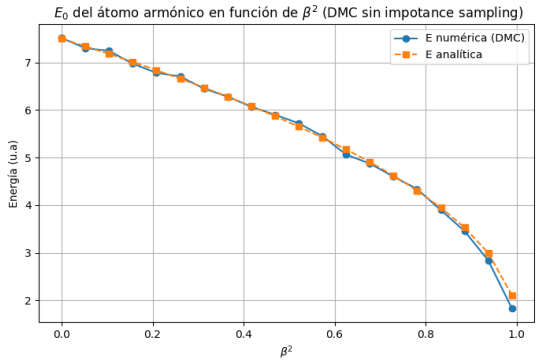
In all cases the ground-state energy decreases monotonically with β^2, reflecting the weakening of the confinement as the interaction strength increases. The small deviations can be attributed to statistical fluctuations in the sampling and, to a lesser extent, to the discretisation error.

For $N = 2$, $N = 5$, and $N = 10$ there is overall agreement between the numerical results and the analytical solution across the full range of β^2. For $N = 20$, however, a more pronounced discrepancy appears, especially at small and intermediate values of β^2, where DMC tends to overestimate the ground-state energy. This deviation originates in the growth of configuration space with particle number, which makes efficient sampling by free diffusion increasingly difficult. A significant fraction of walkers explore low-probability regions, raising the variance. The effects of finite population control and the discrete time step are also amplified in larger systems.

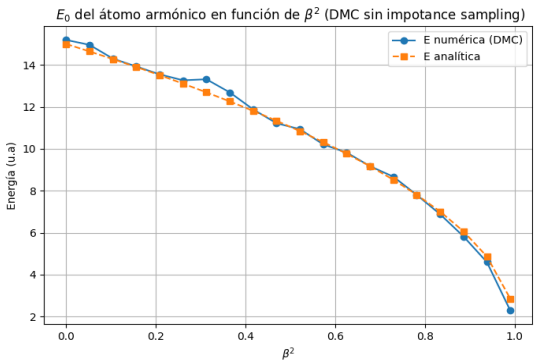
This motivates the use of importance sampling to obtain accurate results for systems with a large number of particles or strong interactions.



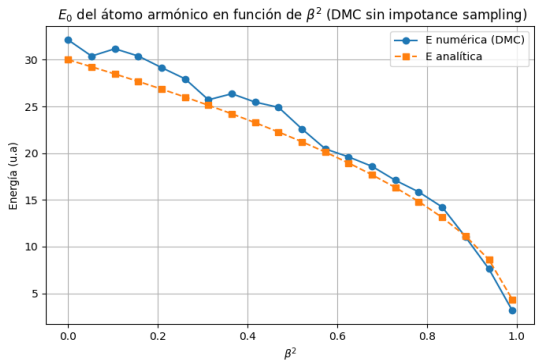
(a) $N = 2$.



(b) $N = 5$.



(c) $N = 10$.



(d) $N = 20$.

Figure 1: Ground-state energy as a function of β^2 for different particle numbers. Atomic units; time step $\Delta\tau = 0.001$; 10^4 steps; target population $2 \cdot 10^3$ walkers. Thermalisation: first $3 \cdot 10^3$ steps discarded. Results averaged over 10^3 realisations.

2.2 Importance Sampling

As the trial wavefunction we take

$$|\Psi_\alpha\rangle = \left(\frac{\alpha}{\pi}\right)^{\frac{3}{4}N} \exp\left(-\frac{\alpha}{2}(\mathbf{r}_1^2 + \cdots + \mathbf{r}_N^2)\right), \tag{2.3}$$

which is the exact ground state of the non-interacting harmonic oscillator with frequency $\omega = \alpha$. Walkers are also initialised from this distribution. The drift force is

$$\mathbf{F}(\mathbf{r}) = 2D\nabla \ln \Psi_\alpha(\mathbf{r}) = -2D\alpha\,(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N), \tag{2.4}$$

which is a restoring force directed towards the origin of configuration space. The Langevin equation becomes

$$\mathbf{r}_{n+1} = \mathbf{r}_n - 2D\alpha\,\mathbf{r}_n\Delta\tau + \sqrt{2D\Delta\tau}\,\boldsymbol{\eta}_n. \tag{2.5}$$

The local energy is

$$
\begin{aligned}
E_L(\mathbf{r}) = \frac{\hat{H}\Psi_T(\mathbf{r})}{\Psi_T(\mathbf{r})} &= \frac{3}{2}N\alpha + \frac{1}{2}(1-\alpha^2)\sum_{k=1}^{N} r_k^2 - \frac{\beta^2}{2N}\sum_{l>k} r_{kl}^2 \\
&= \frac{3}{2}N\alpha + \frac{1}{2}(1-\alpha^2-\beta^2)\sum_{k=1}^{N} r_k^2 + \frac{\beta^2}{2N}\left(\sum_{k=1}^{N} r_k\right)^2 .
\end{aligned}
\tag{2.6}
$$

With this transformation sampling becomes concentrated in the physically relevant regions of configuration space, since the interacting ground state is expected to resemble the non-interacting one.

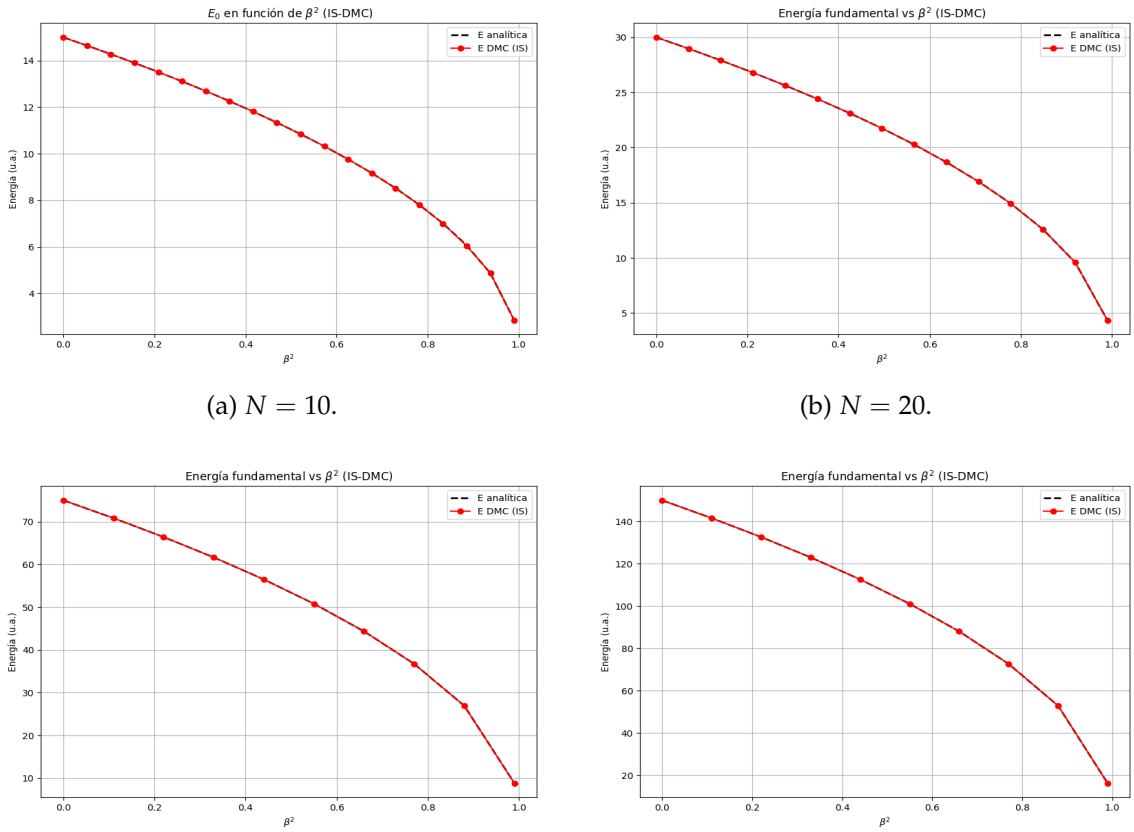
The local energy depends on α, which controls the width of the Gaussian and the localisation of walkers. If Ψ_α were the exact ground state, the local energy would be constant and the variance would vanish. In the non-interacting limit ($\beta = 0$), the exact value $\alpha = 1$ gives a constant $E_L = \frac{3}{2}N$. In the presence of interactions, the β^2 term modifies the effective wavefunction width, requiring an adjustment of α. Setting the coefficient of $\sum_k r_k^2$ in the local energy to zero gives the near-optimal estimate

$$
\alpha_{\text{opt}} \simeq \sqrt{1-\beta^2}. \tag{2.7}
$$

2.2.1 Results

Figure 2 shows the numerical results obtained with IS-DMC described in section 1.2, together with the analytical solution equation 2.2, for several values of N.

In all cases the agreement is excellent, even at large β^2 and large N. The advantage is particularly striking at large particle numbers: an equivalent deterministic calculation sampling m points per dimension would cost $\mathcal{O}(m^{3N})$ operations. For a modest $m = 20$ and $N = 10$ the cost is $20^{30} \sim 10^{39}$; even at 10^{18} operations per second the computation would exceed the age of the universe.



(a) $N = 10$.

(b) $N = 20$.

(c) $N = 50$.

(d) $N = 100$.

Figure 2: Ground-state energy as a function of β^2 for different particle numbers. Atomic units; time step $\Delta\tau = 0.001$; 10^4 steps; target population $2 \cdot 10^3$ walkers. Thermalisation: first $3 \cdot 10^3$ steps discarded. Results averaged over 10^3 realisations.

3 Conclusion

The ground-state energy of a system of bosonic particles under a harmonic potential has been studied numerically as a function of the interaction parameter β^2 and the particle number N, using the Diffusion Monte Carlo method, both in its basic form (pure DMC) and with importance sampling via the Trotter approximation.

Pure DMC shows good agreement with the exact solution for small particle numbers. As N increases, however, the error grows noticeably due to inefficient sampling of configuration space by free diffusion.

Importance sampling with a Gaussian trial wavefunction adds a drift term that guides walkers towards the physically relevant regions and replaces the bare potential with the local energy in the branching step. IS-DMC shows excellent agreement with the analytical solution across the full range of β^2, even for large particle numbers.

References

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